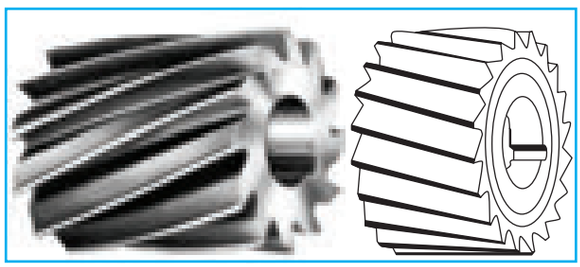
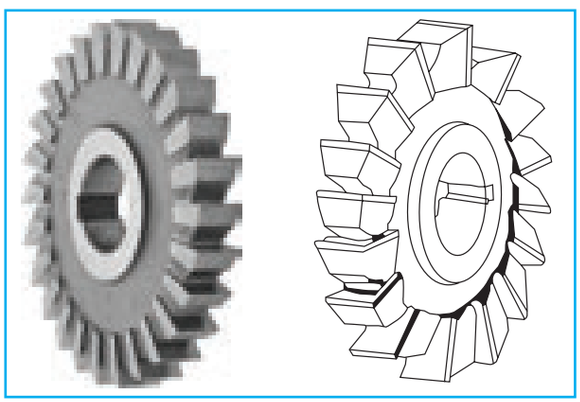
centres. It is also used in shaping machines and slotting machines. While machining gears, spirals, clutches and ratchets this dividing head is used to divide the circumference of the work into any number of equal parts.

## 5.11 Isometric view of milling cutters

Milling cutters are multipoint cutters. Thease cutters are used to remove excess material of given job by milling



Isometric view of milling cutters



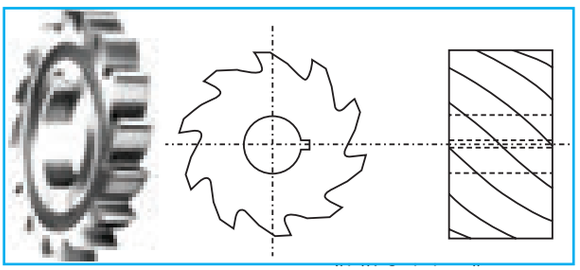
Milling cutters

## 5.12 Standard Milling Cutter

There are different types of milling cutters used in a milling machine. A suitable milling cutter is selected according to the need.

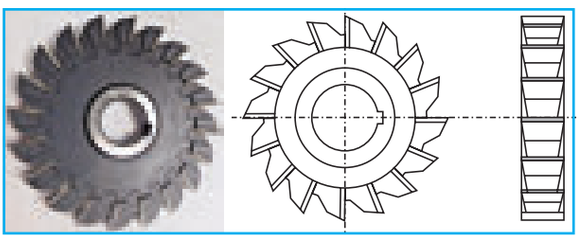
### Plain milling cutter

Plain milling cutters are cylindrical in shape and have teeth on the circumferential surface only. They are used for producing flat surfaces parallel to the axis of rotation of the spindle. The teeth of the cutter may be straight or helical according to the size. If the width of the cutter is more, it is called as slabbing cutter in order to be mounted on the arbor. Plain milling cutters have nicked teeth to break the chips into small pieces. Helical plain milling machine cutters are superior to a straight plain milling cutter.



A plain milling cutter

### Side milling cutter



Side milling cutter

Side milling cutters have teeth on its periphery and also on one or both of its sides. They are intended for removing metal from the sides of the workpiece. There are different types of side milling cutters namely face and side milling cutter, half side milling cutter, staggered teeth side milling cutter and inter locked side milling cutter. Machining is performed by selections a proper milling cutter.

### Metal Slitting saw

It is intended for cutting narrow, deep slots and for parting off operation. The teeth are cut on the circumference of the cutter. The width of the cutter ranges



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